Hyperparameter Optimisation of Vision Transformers (ViTs) to Identify Malignant Colorectal Tumours

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Introduction

Background

Cancer is one of the leading causes of death globally, accounting for an estimated 9.6 million deaths, or 1 in 6 deaths, in 2018 [1]. While there is no universal cure for cancer, research has shown that the cancer burden can effectively be reduced through the early detection of cancer when the disease has not spread [2].

While pretrained models have been developed to handle these tasks and have been largely successful, many limitations still exist with its usage. Firstly, pretrained deep learning models are computationally intensive [3], often requiring multiple GPUs for training [4]. Additionally, pretrained models are often relatively inflexible, especially if the model was trained on a largely different dataset to solve another problem [5].

Rationale

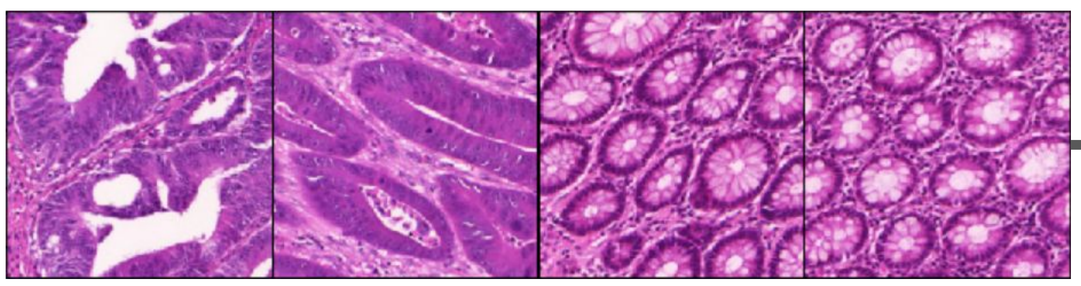
This project hence aims to develop a ViT model from scratch with tuned hyperparameters, trained on a small dataset to identify malignant tumours with reasonable accuracy, aiding in the early detection of cancer through the use of AI.

Aims of the Project

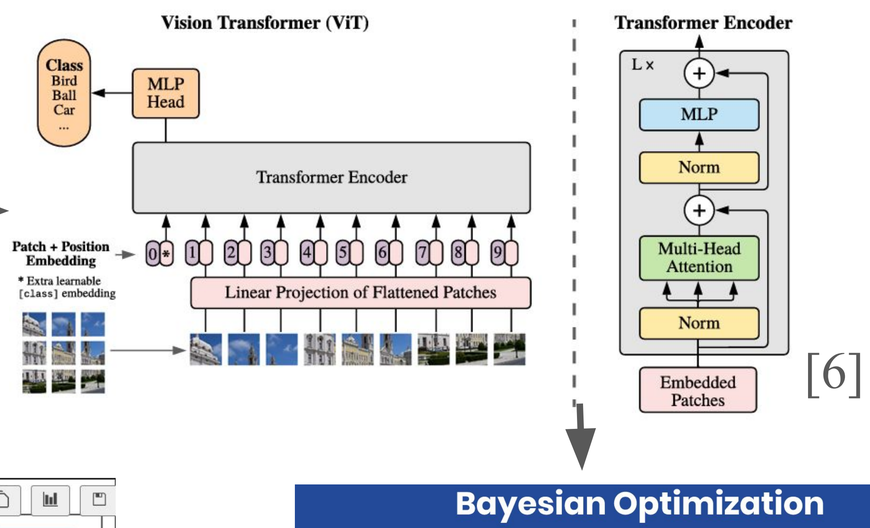
1. Develop a simple and effective ViT model from scratch which can be trained with little computational power.
2. By fine tuning hyperparameters, achieve reasonable accuracy and F1 score on identifying malignant tumours, and compare the results with other models such as CNNs.

Methodology

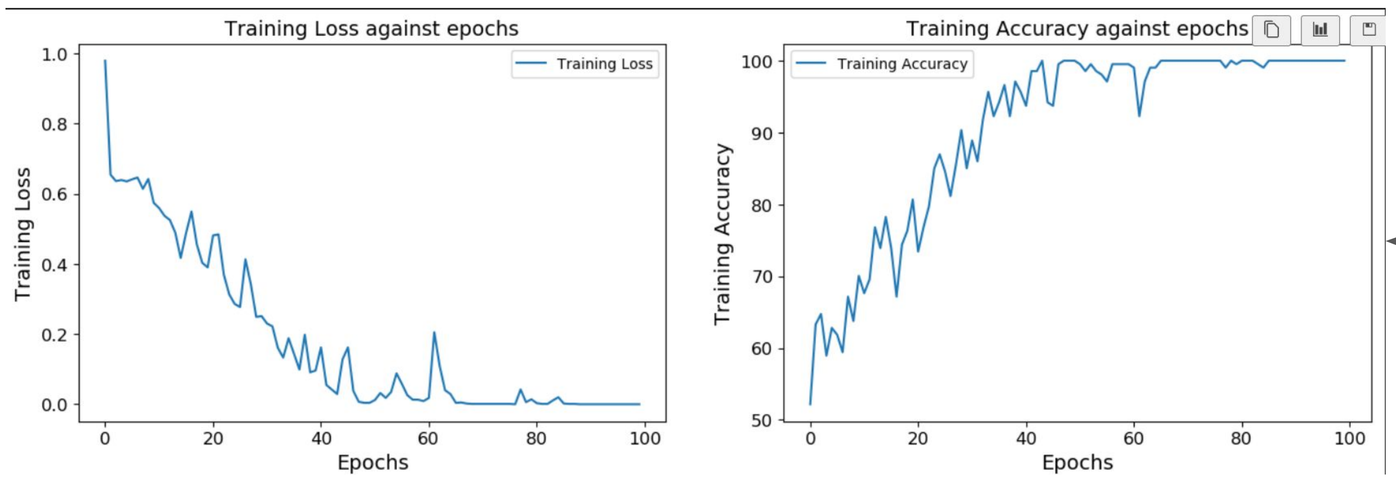
Preparation and augmentation of dataset



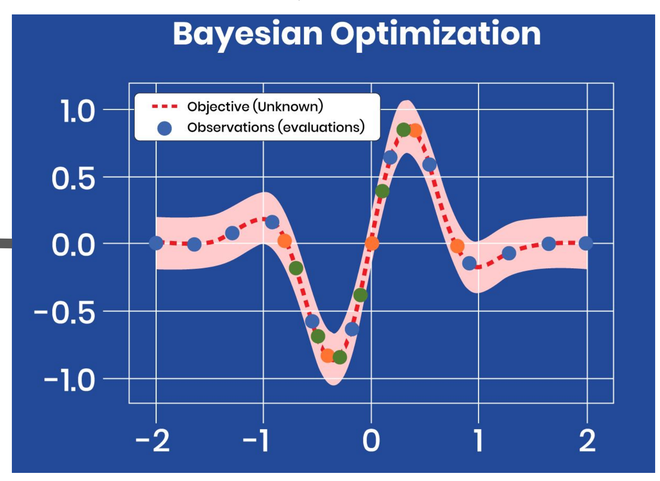
Development of ViT model



[6]



Training Curves

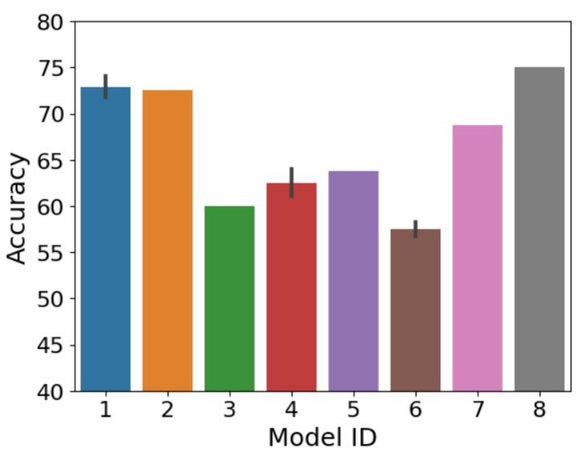


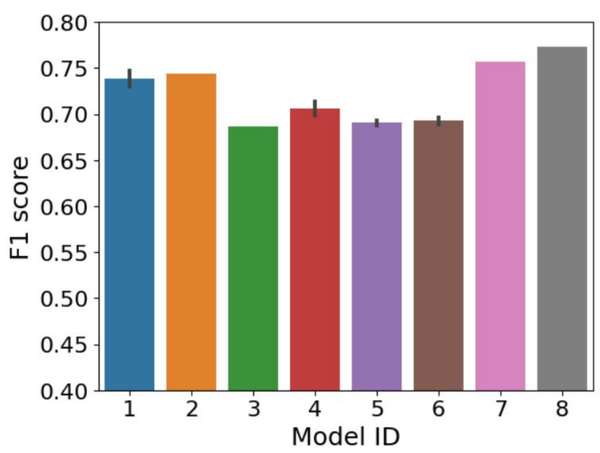
[7]

Optimisation of Hyperparameters

Results

ID	Model	Optimisation	Test Accuracy	F1 score
1	ViT trained from scratch	Grid Search	$72.9 \pm 1.1\%$	0.739 ± 0.009
2	ViT trained from scratch	Bayesian	$72.5 \pm 0.0\%$	0.744 ± 0.000
3	ViT trained from scratch	default	$60.0 \pm 0.0\%$	0.686 ± 0.000
4	CNN trained from scratch	Grid Search	$62.5 \pm 1.4\%$	0.706 ± 0.008
5	CNN trained from scratch	Bayesian	$63.8 \pm 0.0\%$	0.690 ± 0.002
6	CNN trained from scratch	default	$57.5 \pm 0.7\%$	0.693 ± 0.004
7	timm vit_base_patch16_224	default	$68.8 \pm 0.0\%$	0.757 ± 0.000
8	AlexNet CNN	default	$\mathbf{75.0 \pm 0.0\%}$	$\mathbf{0.773 \pm 0.000}$





Evaluation

The performance of the models trained from scratch with tuned hyperparameters are **largely comparable** to that of the pretrained models.

Hyperparameter tuning has **substantially** improved the performance of models, but to a **greater extent** in the **ViT** than the CNN.

Conclusion

Models trained from scratch were relatively **lacklustre** in performance, but the hyperparameter tuning has **significantly improved** their performance, and has shown that models trained from scratch with tuned hyperparameters can also achieve **comparable results** to currently existing pretrained models.

Future Work

More experiments for hyperparameter tuning on tasks which require **multiple classes** could be conducted to determine the **optimal values** of hyperparameters for such tasks.

References

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